Course Information Sheet

Engineering Mathematics (ET1113)

Document No. CRS-0005 · Academic Year 2024/2025 · Semester 1

Course Details

Field	Details
Course ID	5
Course Name	Engineering Mathematics
Course Code	ET1113
Credit Value	3.00
Department	Engineering Technology (Department ID: 2)

Course Description

This course provides the core mathematical toolkit needed by engineering technology students. It reviews differential and integral calculus, introduces linear algebra with matrices and vector spaces, and studies first-order differential equations with engineering applications. Emphasis is placed on formulating mathematical models of physical systems and interpreting solutions in engineering terms. Tutorial sessions apply each technique to problems drawn from mechanics, circuits, and thermal systems.

Learning Outcomes

On successful completion of this course, students will be able to:

- Apply differentiation and integration to engineering problems.
- Solve systems of linear equations using matrix methods.
- Model and solve first-order differential equations arising in engineering systems.
- Interpret mathematical results in the context of physical engineering behaviour.

Assessment and Grading

Component	Weighting	Description
Assignments and Quizzes	20%	Graded coursework issued and completed during the semester
Mid-Semester Test	10%	Written test covering the first half of the course content
Final Examination	70%	End-of-semester written examination

Note: this is a sample document generated for project testing purposes. Assessment weights and descriptive text are illustrative.